Class 9: Candy Mini-Project

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Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA).

Importing candy data

The data is all realted to Halloween candy and is from the 538 website.

```
candy <- read.csv("candy-data.txt", row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
```

```
Air Heads             0       1        0                0       0                 0
Almond Joy            1       0        0                1       0                 0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

> **Q1.** How many different candy types are in this dataset?

```r
nrow(candy)
```

```
[1] 85
```

85 different candy types are in the dataset.

> **Q2.** How many fruity candy types are in the dataset?

```r
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types.

> **Q3.** What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy %>%
  filter(row.names(candy)=="Milky Way") |>
  select(winpercent)
```

```
          winpercent
Milky Way   73.09956
```

The winpercent value of "Milky Way" is 73.09956.

> **Q4.** What is the winpercent value for "Kit Kat"?

```r
candy %>%
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
        winpercent
Kit Kat    76.7686
```

The winpercent value is 76.7686.

> **Q5.** What is the winpercent value for "Tootsie Roll Snack Bars"?

```r
candy %>%
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
                        winpercent
Tootsie Roll Snack Bars    49.6535
```

The winpercent value is 49.6535.

There is a useful 'skim()' function in the **skimr** package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```r
library(skimr)
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> **Q6.** Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

winpercent seems to be on a different scale.

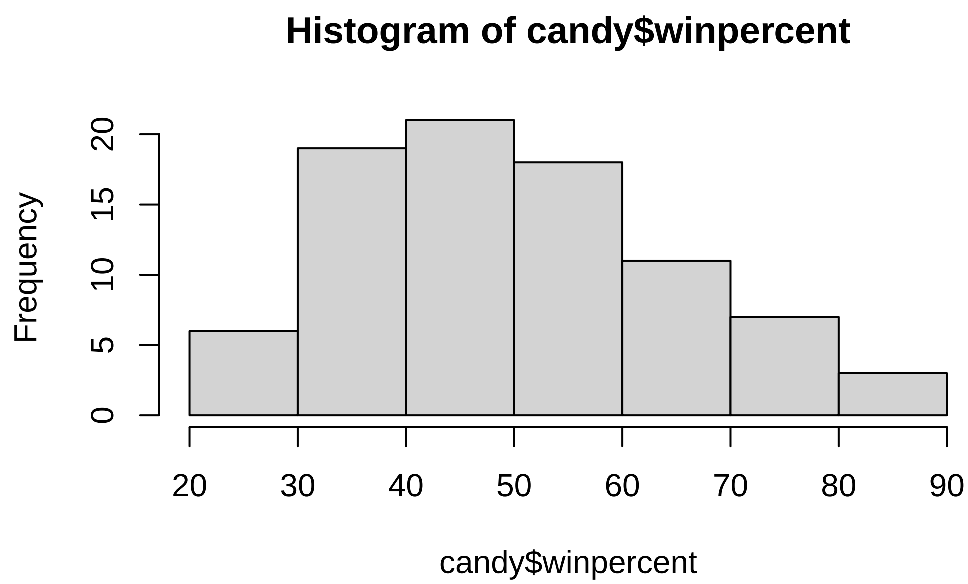
> **Q7.** What do you think a zero and one represent for the candy$chocolate column?

They represent whether the candy is chocolate or not. If the value is zero, then it's not chocolate. If the value is one, it is a chocolate.

Exploratory analysis

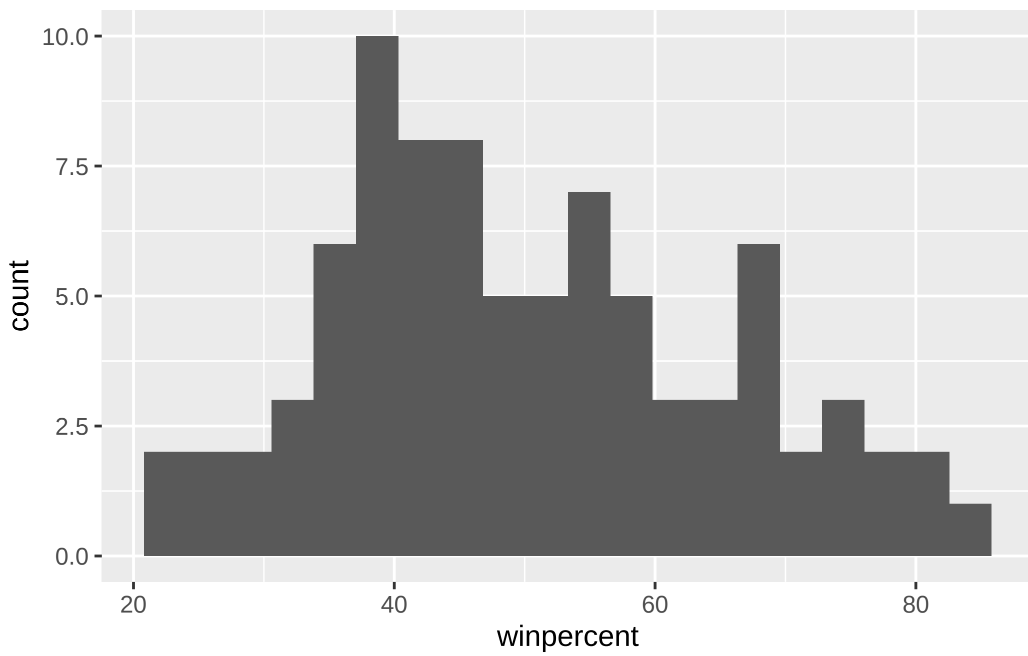
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent)
```




```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins = 20)
```



Q9. Is the distribution of winpercent values symmetrical?

No

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

Below 50%

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```r
fruity.ind <- as.logical(candy$fruity)
fruity.candy <- candy[fruity.ind, ]
fruity.win <- fruity.candy$winpercent
mean(fruity.win)
```

```
[1] 44.11974
```

Chocolate candy is ranked higher on average than fruity candy.

> **Q12.** Is this difference statistically significant?

```r
t.test(choc.win, fruity.win)
```

```

	Welch Two Sample t-test

data:  choc.win and fruity.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

Yes, this difference is statistically significant.

Overall Candy Rankings

> **Q13.** What are the five least liked candy types in this set?

```r
x <- c(5, 10, 1)
sort(x)
```

```
[1]  1  5 10
```

```r
order(x)
```

```
[1] 3 1 2
```

```r
candy[order(candy$winpercent), ]
```

```
                            chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                           0      1       0              0      0
Boston Baked Beans                  0      0       0              1      0
Chiclets                            0      1       0              0      0
Super Bubble                        0      1       0              0      0
Jawbusters                          0      1       0              0      0
Root Beer Barrels                   0      0       0              0      0
Sugar Daddy                         0      0       1              0      0
One dime                            0      0       0              0      0
Sugar Babies                        0      0       1              0      0
Haribo Happy Cola                   0      0       0              0      0
Caramel Apple Pops                  0      1       1              0      0
Strawberry bon bons                 0      1       0              0      0
Sixlets                             1      0       0              0      0
Ring pop                            0      1       0              0      0
Chewey Lemonhead Fruit Mix          0      1       0              0      0
Red vines                           0      1       0              0      0
Pixie Sticks                        0      0       0              0      0
Nestle Smarties                     1      0       0              0      0
Candy Corn                          0      0       0              0      0
Charleston Chew                     1      0       0              0      1
Warheads                            0      1       0              0      0
Lemonhead                           0      1       0              0      0
Fun Dip                             0      1       0              0      0
Now & Later                         0      1       0              0      0
Dum Dums                            0      1       0              0      0
Pop Rocks                           0      1       0              0      0
Laffy Taffy                         0      1       0              0      0
Werther's Original Caramel          0      0       1              0      0
Haribo Twin Snakes                  0      1       0              0      0
Dots                                0      1       0              0      0
Runts                               0      1       0              0      0
Tootsie Roll Juniors                1      0       0              0      0
Fruit Chews                         0      1       0              0      0
Welch's Fruit Snacks                0      1       0              0      0
Twizzlers                           0      1       0              0      0
Tootsie Roll Midgies                1      0       0              0      0
Smarties candy                      0      1       0              0      0
One quarter                         0      0       0              0      0
Payday                              0      0       0              1      1
```

Mike & Ike	0	1	0	0	0
Gobstopper	0	1	0	0	0
Trolli Sour Bites	0	1	0	0	0
Mounds	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Whoppers	1	0	0	0	0
Tootsie Roll Snack Bars	1	0	0	0	0
Almond Joy	1	0	0	1	0
Haribo Sour Bears	0	1	0	0	0
Air Heads	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Lifesavers big ring gummies	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Swedish Fish	0	1	0	0	0
Milk Duds	1	0	1	0	0
Skittles wildberry	0	1	0	0	0
Nerds	0	1	0	0	0
Hershey's Kisses	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
Baby Ruth	1	0	1	1	1
Haribo Gold Bears	0	1	0	0	0
Junior Mints	1	0	0	0	0
Hershey's Special Dark	1	0	0	0	0
Snickers Crisper	1	0	1	1	0
Sour Patch Kids	0	1	0	0	0
Milky Way Midnight	1	0	1	0	1
Hershey's Krackel	1	0	0	0	0
Skittles original	0	1	0	0	0
Milky Way Simply Caramel	1	0	1	0	0
Rolo	1	0	1	0	0
Nestle Crunch	1	0	0	0	0
M&M's	1	0	0	0	0
100 Grand	1	0	1	0	0
Starburst	0	1	0	0	0
3 Musketeers	1	0	0	0	1
Peanut M&Ms	1	0	0	1	0
Nestle Butterfinger	1	0	0	1	0
Peanut butter M&M's	1	0	0	1	0
Reese's stuffed with pieces	1	0	0	1	0
Milky Way	1	0	1	0	1
Reese's pieces	1	0	0	1	0
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0

Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0

	crispedricewafer	hard	bar	pluribus	sugarpercent
Nik L Nip	0	0	0	1	0.197
Boston Baked Beans	0	0	0	1	0.313
Chiclets	0	0	0	1	0.046
Super Bubble	0	0	0	0	0.162
Jawbusters	0	1	0	1	0.093
Root Beer Barrels	0	1	0	1	0.732
Sugar Daddy	0	0	0	0	0.418
One dime	0	0	0	0	0.011
Sugar Babies	0	0	0	1	0.965
Haribo Happy Cola	0	0	0	1	0.465
Caramel Apple Pops	0	0	0	0	0.604
Strawberry bon bons	0	1	0	1	0.569
Sixlets	0	0	0	1	0.220
Ring pop	0	1	0	0	0.732
Chewey Lemonhead Fruit Mix	0	0	0	1	0.732
Red vines	0	0	0	1	0.581
Pixie Sticks	0	0	0	1	0.093
Nestle Smarties	0	0	0	1	0.267
Candy Corn	0	0	0	1	0.906
Charleston Chew	0	0	1	0	0.604
Warheads	0	1	0	0	0.093
Lemonhead	0	1	0	0	0.046
Fun Dip	0	1	0	0	0.732
Now & Later	0	0	0	1	0.220
Dum Dums	0	1	0	0	0.732
Pop Rocks	0	1	0	1	0.604
Laffy Taffy	0	0	0	0	0.220
Werther's Original Caramel	0	1	0	0	0.186
Haribo Twin Snakes	0	0	0	1	0.465
Dots	0	0	0	1	0.732
Runts	0	1	0	1	0.872
Tootsie Roll Juniors	0	0	0	0	0.313
Fruit Chews	0	0	0	1	0.127
Welch's Fruit Snacks	0	0	0	1	0.313
Twizzlers	0	0	0	0	0.220
Tootsie Roll Midgies	0	0	0	1	0.174
Smarties candy	0	1	0	1	0.267
One quarter	0	0	0	0	0.011
Payday	0	0	1	0	0.465

Mike & Ike	0	0	0	1	0.872
Gobstopper	0	1	0	1	0.906
Trolli Sour Bites	0	0	0	1	0.313
Mounds	0	0	1	0	0.313
Tootsie Pop	0	1	0	0	0.604
Whoppers	1	0	0	1	0.872
Tootsie Roll Snack Bars	0	0	1	0	0.465
Almond Joy	0	0	1	0	0.465
Haribo Sour Bears	0	0	0	1	0.465
Air Heads	0	0	0	0	0.906
Sour Patch Tricksters	0	0	0	1	0.069
Lifesavers big ring gummies	0	0	0	0	0.267
Mr Good Bar	0	0	1	0	0.313
Swedish Fish	0	0	0	1	0.604
Milk Duds	0	0	0	1	0.302
Skittles wildberry	0	0	0	1	0.941
Nerds	0	1	0	1	0.848
Hershey's Kisses	0	0	0	1	0.127
Hershey's Milk Chocolate	0	0	1	0	0.430
Baby Ruth	0	0	1	0	0.604
Haribo Gold Bears	0	0	0	1	0.465
Junior Mints	0	0	0	1	0.197
Hershey's Special Dark	0	0	1	0	0.430
Snickers Crisper	1	0	1	0	0.604
Sour Patch Kids	0	0	0	1	0.069
Milky Way Midnight	0	0	1	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Skittles original	0	0	0	1	0.941
Milky Way Simply Caramel	0	0	1	0	0.965
Rolo	0	0	0	1	0.860
Nestle Crunch	1	0	1	0	0.313
M&M's	0	0	0	1	0.825
100 Grand	1	0	1	0	0.732
Starburst	0	0	0	1	0.151
3 Musketeers	0	0	1	0	0.604
Peanut M&Ms	0	0	0	1	0.593
Nestle Butterfinger	0	0	1	0	0.604
Peanut butter M&M's	0	0	0	1	0.825
Reese's stuffed with pieces	0	0	0	0	0.988
Milky Way	0	0	1	0	0.604
Reese's pieces	0	0	0	1	0.406
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313

```
Twix                            1     0    1         0          0.546
Reese's Miniatures              0     0    0         0          0.034
Reese's Peanut Butter cup       0     0    0         0          0.720
                            pricepercent winpercent
Nik L Nip                          0.976   22.44534
Boston Baked Beans                 0.511   23.41782
Chiclets                           0.325   24.52499
Super Bubble                       0.116   27.30386
Jawbusters                         0.511   28.12744
Root Beer Barrels                  0.069   29.70369
Sugar Daddy                        0.325   32.23100
One dime                           0.116   32.26109
Sugar Babies                       0.767   33.43755
Haribo Happy Cola                  0.465   34.15896
Caramel Apple Pops                 0.325   34.51768
Strawberry bon bons                0.058   34.57899
Sixlets                            0.081   34.72200
Ring pop                           0.965   35.29076
Chewey Lemonhead Fruit Mix         0.511   36.01763
Red vines                          0.116   37.34852
Pixie Sticks                       0.023   37.72234
Nestle Smarties                    0.976   37.88719
Candy Corn                         0.325   38.01096
Charleston Chew                    0.511   38.97504
Warheads                           0.116   39.01190
Lemonhead                          0.104   39.14106
Fun Dip                            0.325   39.18550
Now & Later                        0.325   39.44680
Dum Dums                           0.034   39.46056
Pop Rocks                          0.837   41.26551
Laffy Taffy                        0.116   41.38956
Werther's Original Caramel         0.267   41.90431
Haribo Twin Snakes                 0.465   42.17877
Dots                               0.511   42.27208
Runts                              0.279   42.84914
Tootsie Roll Juniors               0.511   43.06890
Fruit Chews                        0.034   43.08892
Welch's Fruit Snacks               0.313   44.37552
Twizzlers                          0.116   45.46628
Tootsie Roll Midgies               0.011   45.73675
Smarties candy                     0.116   45.99583
One quarter                        0.511   46.11650
Payday                             0.767   46.29660
```

Mike & Ike	0.325	46.41172
Gobstopper	0.453	46.78335
Trolli Sour Bites	0.255	47.17323
Mounds	0.860	47.82975
Tootsie Pop	0.325	48.98265
Whoppers	0.848	49.52411
Tootsie Roll Snack Bars	0.325	49.65350
Almond Joy	0.767	50.34755
Haribo Sour Bears	0.465	51.41243
Air Heads	0.511	52.34146
Sour Patch Tricksters	0.116	52.82595
Lifesavers big ring gummies	0.279	52.91139
Mr Good Bar	0.918	54.52645
Swedish Fish	0.755	54.86111
Milk Duds	0.511	55.06407
Skittles wildberry	0.220	55.10370
Nerds	0.325	55.35405
Hershey's Kisses	0.093	55.37545
Hershey's Milk Chocolate	0.918	56.49050
Baby Ruth	0.767	56.91455
Haribo Gold Bears	0.465	57.11974
Junior Mints	0.511	57.21925
Hershey's Special Dark	0.918	59.23612
Snickers Crisper	0.651	59.52925
Sour Patch Kids	0.116	59.86400
Milky Way Midnight	0.441	60.80070
Hershey's Krackel	0.918	62.28448
Skittles original	0.220	63.08514
Milky Way Simply Caramel	0.860	64.35334
Rolo	0.860	65.71629
Nestle Crunch	0.767	66.47068
M&M's	0.651	66.57458
100 Grand	0.860	66.97173
Starburst	0.220	67.03763
3 Musketeers	0.511	67.60294
Peanut M&Ms	0.651	69.48379
Nestle Butterfinger	0.767	70.73564
Peanut butter M&M's	0.651	71.46505
Reese's stuffed with pieces	0.651	72.88790
Milky Way	0.651	73.09956
Reese's pieces	0.651	73.43499
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860

```
Twix                               0.906   81.64291
Reese's Miniatures                 0.279   81.86626
Reese's Peanut Butter cup          0.651   84.18029
```

Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbuster.

Q14. What are the top 5 all time favorite candy types out of this set?

```r
candy[order(candy$winpercent, decreasing = TRUE), ]
```

```
                            chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup           1      0       0              1      0
Reese's Miniatures                  1      0       0              1      0
Twix                                1      0       1              0      0
Kit Kat                             1      0       0              0      0
Snickers                            1      0       1              1      1
Reese's pieces                      1      0       0              1      0
Milky Way                           1      0       1              0      1
Reese's stuffed with pieces         1      0       0              1      0
Peanut butter M&M's                 1      0       0              1      0
Nestle Butterfinger                 1      0       0              1      0
Peanut M&Ms                         1      0       0              1      0
3 Musketeers                        1      0       0              0      1
Starburst                           0      1       0              0      0
100 Grand                           1      0       1              0      0
M&M's                               1      0       0              0      0
Nestle Crunch                       1      0       0              0      0
Rolo                                1      0       1              0      0
Milky Way Simply Caramel            1      0       1              0      0
Skittles original                   0      1       0              0      0
Hershey's Krackel                   1      0       0              0      0
Milky Way Midnight                  1      0       1              0      1
Sour Patch Kids                     0      1       0              0      0
Snickers Crisper                    1      0       1              1      0
Hershey's Special Dark              1      0       0              0      0
Junior Mints                        1      0       0              0      0
Haribo Gold Bears                   0      1       0              0      0
Baby Ruth                           1      0       1              1      1
Hershey's Milk Chocolate            1      0       0              0      0
Hershey's Kisses                    1      0       0              0      0
Nerds                               0      1       0              0      0
Skittles wildberry                  0      1       0              0      0
```

Milk Duds	1	0	1	0	0
Swedish Fish	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Lifesavers big ring gummies	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Air Heads	0	1	0	0	0
Haribo Sour Bears	0	1	0	0	0
Almond Joy	1	0	0	1	0
Tootsie Roll Snack Bars	1	0	0	0	0
Whoppers	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Mounds	1	0	0	0	0
Trolli Sour Bites	0	1	0	0	0
Gobstopper	0	1	0	0	0
Mike & Ike	0	1	0	0	0
Payday	0	0	0	1	1
One quarter	0	0	0	0	0
Smarties candy	0	1	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Twizzlers	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
Fruit Chews	0	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0
Runts	0	1	0	0	0
Dots	0	1	0	0	0
Haribo Twin Snakes	0	1	0	0	0
Werther's Original Caramel	0	0	1	0	0
Laffy Taffy	0	1	0	0	0
Pop Rocks	0	1	0	0	0
Dum Dums	0	1	0	0	0
Now & Later	0	1	0	0	0
Fun Dip	0	1	0	0	0
Lemonhead	0	1	0	0	0
Warheads	0	1	0	0	0
Charleston Chew	1	0	0	0	1
Candy Corn	0	0	0	0	0
Nestle Smarties	1	0	0	0	0
Pixie Sticks	0	0	0	0	0
Red vines	0	1	0	0	0
Chewey Lemonhead Fruit Mix	0	1	0	0	0
Ring pop	0	1	0	0	0
Sixlets	1	0	0	0	0
Strawberry bon bons	0	1	0	0	0

Caramel Apple Pops	0	1	1	0	0
Haribo Happy Cola	0	0	0	0	0
Sugar Babies	0	0	1	0	0
One dime	0	0	0	0	0
Sugar Daddy	0	0	1	0	0
Root Beer Barrels	0	0	0	0	0
Jawbusters	0	1	0	0	0
Super Bubble	0	1	0	0	0
Chiclets	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Nik L Nip	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent
Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546
Reese's pieces	0	0	0	1	0.406
Milky Way	0	0	1	0	0.604
Reese's stuffed with pieces	0	0	0	0	0.988
Peanut butter M&M's	0	0	0	1	0.825
Nestle Butterfinger	0	0	1	0	0.604
Peanut M&Ms	0	0	0	1	0.593
3 Musketeers	0	0	1	0	0.604
Starburst	0	0	0	1	0.151
100 Grand	1	0	1	0	0.732
M&M's	0	0	0	1	0.825
Nestle Crunch	1	0	1	0	0.313
Rolo	0	0	0	1	0.860
Milky Way Simply Caramel	0	0	1	0	0.965
Skittles original	0	0	0	1	0.941
Hershey's Krackel	1	0	1	0	0.430
Milky Way Midnight	0	0	1	0	0.732
Sour Patch Kids	0	0	0	1	0.069
Snickers Crisper	1	0	1	0	0.604
Hershey's Special Dark	0	0	1	0	0.430
Junior Mints	0	0	0	1	0.197
Haribo Gold Bears	0	0	0	1	0.465
Baby Ruth	0	0	1	0	0.604
Hershey's Milk Chocolate	0	0	1	0	0.430
Hershey's Kisses	0	0	0	1	0.127
Nerds	0	1	0	1	0.848
Skittles wildberry	0	0	0	1	0.941

Milk Duds	0	0	0	1	0.302
Swedish Fish	0	0	0	1	0.604
Mr Good Bar	0	0	1	0	0.313
Lifesavers big ring gummies	0	0	0	0	0.267
Sour Patch Tricksters	0	0	0	1	0.069
Air Heads	0	0	0	0	0.906
Haribo Sour Bears	0	0	0	1	0.465
Almond Joy	0	0	1	0	0.465
Tootsie Roll Snack Bars	0	0	1	0	0.465
Whoppers	1	0	0	1	0.872
Tootsie Pop	0	1	0	0	0.604
Mounds	0	0	1	0	0.313
Trolli Sour Bites	0	0	0	1	0.313
Gobstopper	0	1	0	1	0.906
Mike & Ike	0	0	0	1	0.872
Payday	0	0	1	0	0.465
One quarter	0	0	0	0	0.011
Smarties candy	0	1	0	1	0.267
Tootsie Roll Midgies	0	0	0	1	0.174
Twizzlers	0	0	0	0	0.220
Welch's Fruit Snacks	0	0	0	1	0.313
Fruit Chews	0	0	0	1	0.127
Tootsie Roll Juniors	0	0	0	0	0.313
Runts	0	1	0	1	0.872
Dots	0	0	0	1	0.732
Haribo Twin Snakes	0	0	0	1	0.465
Werther's Original Caramel	0	1	0	0	0.186
Laffy Taffy	0	0	0	0	0.220
Pop Rocks	0	1	0	1	0.604
Dum Dums	0	1	0	0	0.732
Now & Later	0	0	0	1	0.220
Fun Dip	0	1	0	0	0.732
Lemonhead	0	1	0	0	0.046
Warheads	0	1	0	0	0.093
Charleston Chew	0	0	1	0	0.604
Candy Corn	0	0	0	1	0.906
Nestle Smarties	0	0	0	1	0.267
Pixie Sticks	0	0	0	1	0.093
Red vines	0	0	0	1	0.581
Chewey Lemonhead Fruit Mix	0	0	0	1	0.732
Ring pop	0	1	0	0	0.732
Sixlets	0	0	0	1	0.220
Strawberry bon bons	0	1	0	1	0.569

Caramel Apple Pops	0	0	0	0	0.604
Haribo Happy Cola	0	0	0	1	0.465
Sugar Babies	0	0	0	1	0.965
One dime	0	0	0	0	0.011
Sugar Daddy	0	0	0	0	0.418
Root Beer Barrels	0	1	0	1	0.732
Jawbusters	0	1	0	1	0.093
Super Bubble	0	0	0	0	0.162
Chiclets	0	0	0	1	0.046
Boston Baked Beans	0	0	0	1	0.313
Nik L Nip	0	0	0	1	0.197

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378
Reese's pieces	0.651	73.43499
Milky Way	0.651	73.09956
Reese's stuffed with pieces	0.651	72.88790
Peanut butter M&M's	0.651	71.46505
Nestle Butterfinger	0.767	70.73564
Peanut M&Ms	0.651	69.48379
3 Musketeers	0.511	67.60294
Starburst	0.220	67.03763
100 Grand	0.860	66.97173
M&M's	0.651	66.57458
Nestle Crunch	0.767	66.47068
Rolo	0.860	65.71629
Milky Way Simply Caramel	0.860	64.35334
Skittles original	0.220	63.08514
Hershey's Krackel	0.918	62.28448
Milky Way Midnight	0.441	60.80070
Sour Patch Kids	0.116	59.86400
Snickers Crisper	0.651	59.52925
Hershey's Special Dark	0.918	59.23612
Junior Mints	0.511	57.21925
Haribo Gold Bears	0.465	57.11974
Baby Ruth	0.767	56.91455
Hershey's Milk Chocolate	0.918	56.49050
Hershey's Kisses	0.093	55.37545
Nerds	0.325	55.35405
Skittles wildberry	0.220	55.10370

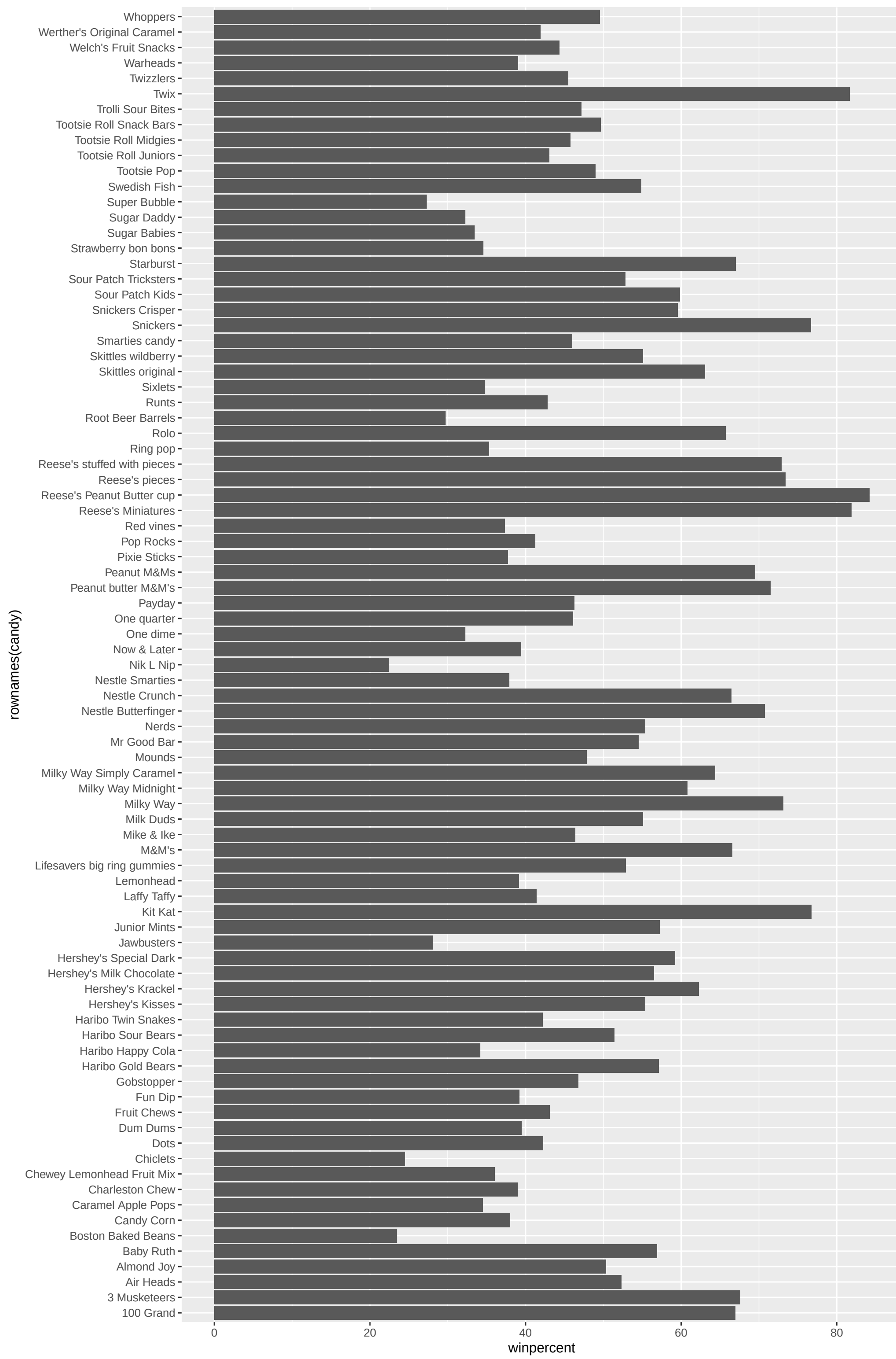
Milk Duds	0.511	55.06407
Swedish Fish	0.755	54.86111
Mr Good Bar	0.918	54.52645
Lifesavers big ring gummies	0.279	52.91139
Sour Patch Tricksters	0.116	52.82595
Air Heads	0.511	52.34146
Haribo Sour Bears	0.465	51.41243
Almond Joy	0.767	50.34755
Tootsie Roll Snack Bars	0.325	49.65350
Whoppers	0.848	49.52411
Tootsie Pop	0.325	48.98265
Mounds	0.860	47.82975
Trolli Sour Bites	0.255	47.17323
Gobstopper	0.453	46.78335
Mike & Ike	0.325	46.41172
Payday	0.767	46.29660
One quarter	0.511	46.11650
Smarties candy	0.116	45.99583
Tootsie Roll Midgies	0.011	45.73675
Twizzlers	0.116	45.46628
Welch's Fruit Snacks	0.313	44.37552
Fruit Chews	0.034	43.08892
Tootsie Roll Juniors	0.511	43.06890
Runts	0.279	42.84914
Dots	0.511	42.27208
Haribo Twin Snakes	0.465	42.17877
Werther's Original Caramel	0.267	41.90431
Laffy Taffy	0.116	41.38956
Pop Rocks	0.837	41.26551
Dum Dums	0.034	39.46056
Now & Later	0.325	39.44680
Fun Dip	0.325	39.18550
Lemonhead	0.104	39.14106
Warheads	0.116	39.01190
Charleston Chew	0.511	38.97504
Candy Corn	0.325	38.01096
Nestle Smarties	0.976	37.88719
Pixie Sticks	0.023	37.72234
Red vines	0.116	37.34852
Chewey Lemonhead Fruit Mix	0.511	36.01763
Ring pop	0.965	35.29076
Sixlets	0.081	34.72200
Strawberry bon bons	0.058	34.57899

```
Caramel Apple Pops                  0.325   34.51768
Haribo Happy Cola                   0.465   34.15896
Sugar Babies                        0.767   33.43755
One dime                            0.116   32.26109
Sugar Daddy                         0.325   32.23100
Root Beer Barrels                   0.069   29.70369
Jawbusters                          0.511   28.12744
Super Bubble                        0.116   27.30386
Chiclets                            0.325   24.52499
Boston Baked Beans                  0.511   23.41782
Nik L Nip                           0.976   22.44534
```

Reese's Peanut Butter Cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

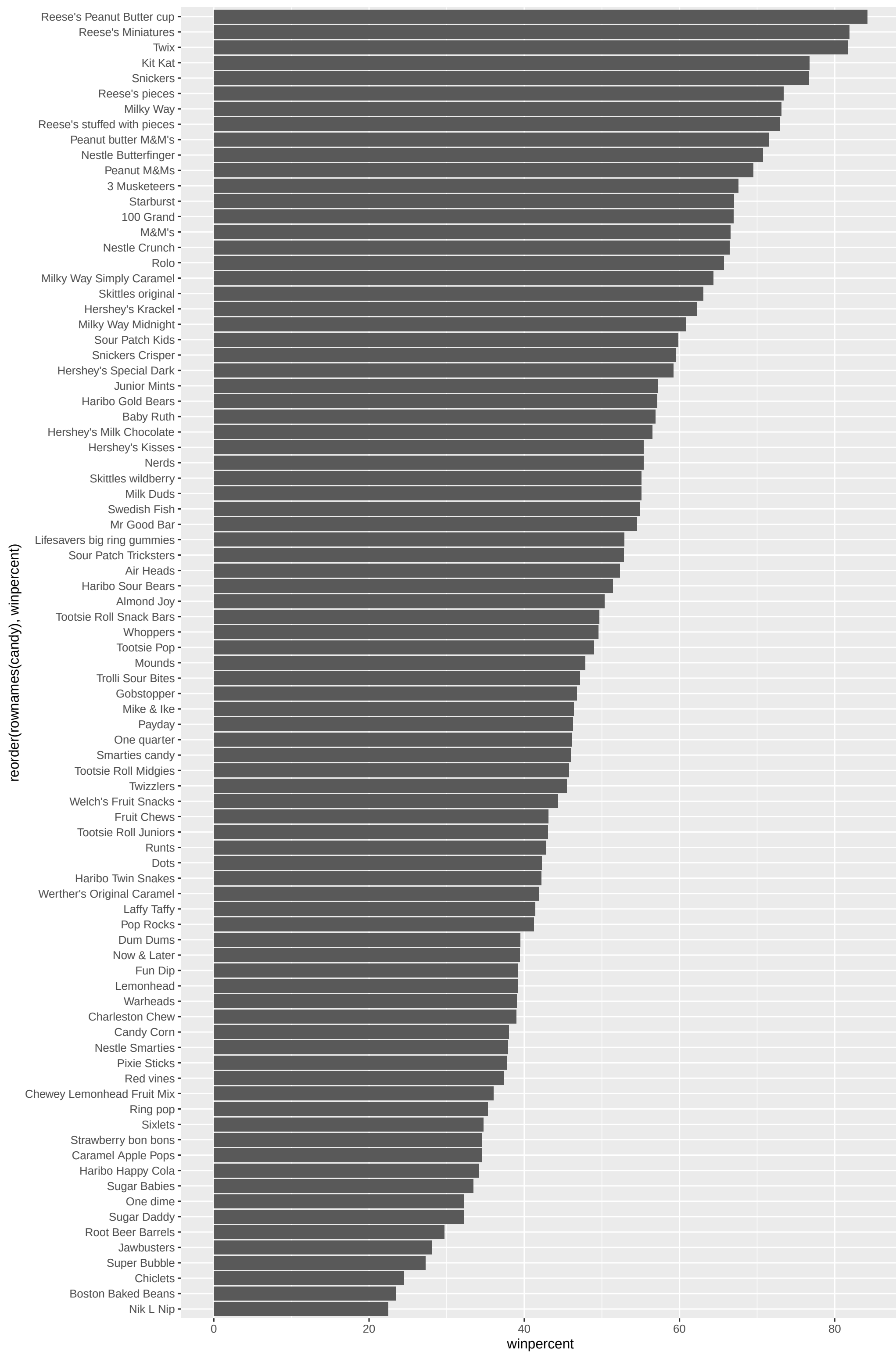
Q15. Make a first barplot of candy ranking based on winpercent values.

```r
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```


Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
rownames(candy)
0
20
40
60
80
winpercent

Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```

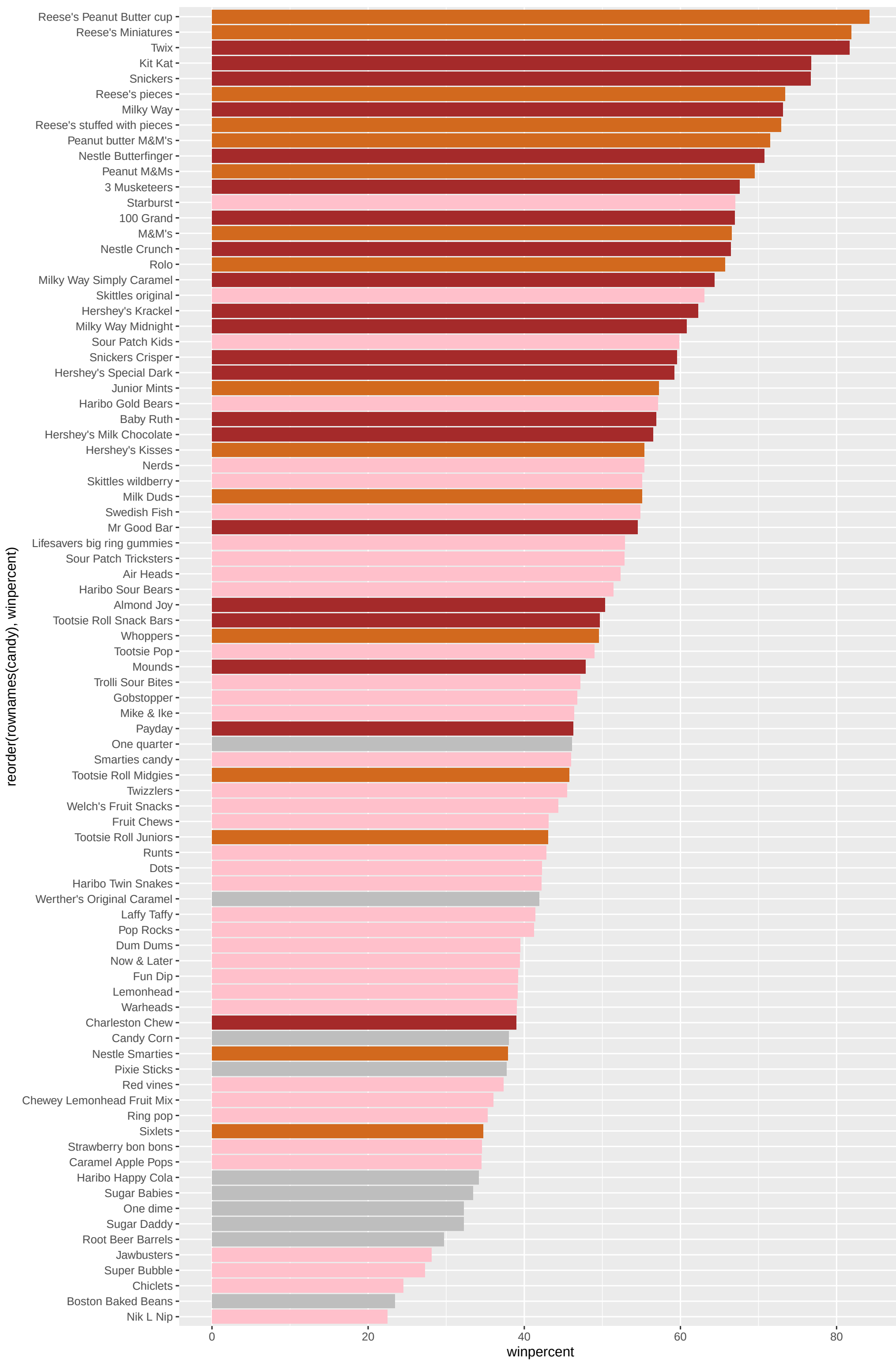

Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

```
mycols <- rep("gray", nrow(candy))

## chocolate candy in chocolate color
mycols[as.logical(candy$chocolate)] <- "chocolate"
## candy bars in brown
mycols[as.logical(candy$bar)] <- "brown"
## fruity candy in pink
mycols[as.logical(candy$fruity)] <- "pink"

ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent)) +
        geom_col(fill=mycols)
```


reorder(rownames(candy), winpercent)
winpercent
0
20
40
60
80
Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip

Sixlets is the worst ranked chocolate candy.

Q18. What is the best ranked fruity candy?

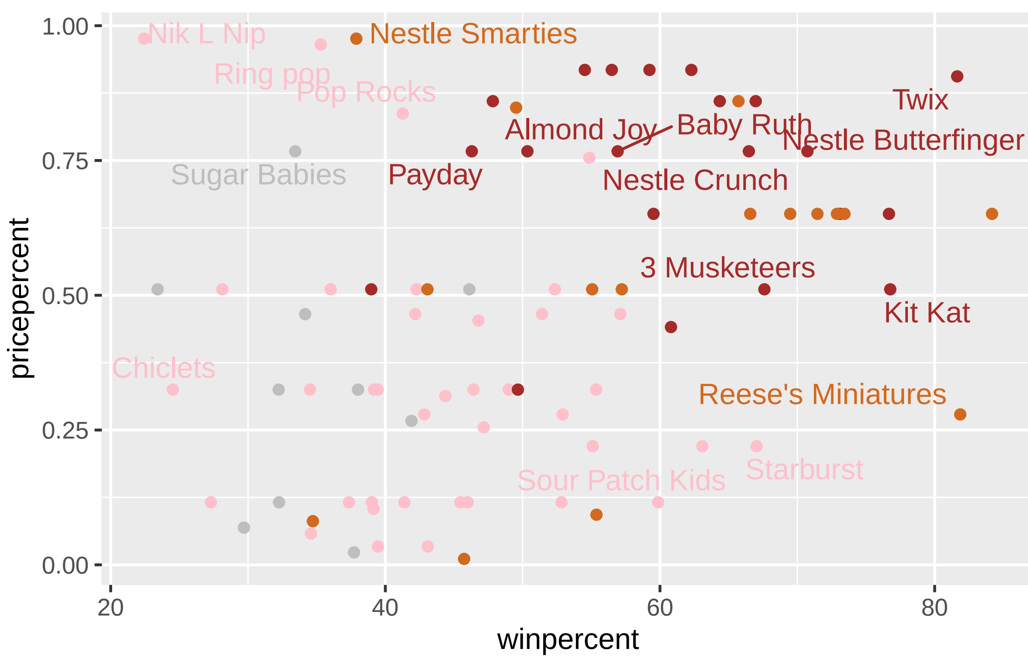
Starburst is the best ranked fruity candy.

Taking a look at pricepercent

Make a plot of winprcent vs pricepercent

```
library(ggrepel)

ggplot(candy) +
  aes(winpercent,
      pricepercent,
      label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, max.overlaps=7)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Based on the figure above, "Reese's Miniatures" is the highest ranked in terms of winpercent for the least money.

> **Q20.** What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

From the figure, I can tell that Nik L Nip, Nestle Smarties, and Ring pop are the most expensive. I can't clearly tell which ones are the next most expensive out of these: Mr Good Bar, Hershey's Milk Chocolate, Hershey's Special Dark, or Hershey's Krackel. Out of those, Nik L Nip is the least popular.

Exploring the correlation structure

We can calculate the pair-wise correlation of all our columns.

```r
cij <- cor(candy)
cij
```

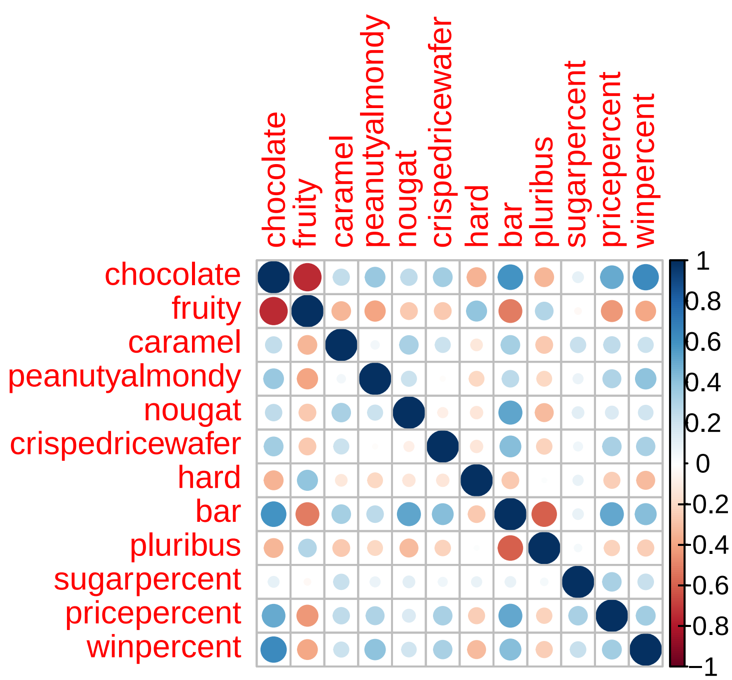
```
                  chocolate      fruity     caramel peanutyalmondy      nougat
chocolate         1.0000000 -0.74172106  0.24987535     0.37782357  0.25489183
fruity           -0.7417211  1.00000000 -0.33548538    -0.39928014 -0.26936712
caramel           0.2498753 -0.33548538  1.00000000     0.05935614  0.32849280
peanutyalmondy    0.3778236 -0.39928014  0.05935614     1.00000000  0.21311310
nougat            0.2548918 -0.26936712  0.32849280     0.21311310  1.00000000
crispedricewafer  0.3412098 -0.26936712  0.21311310    -0.01764631 -0.08974359
hard             -0.3441769  0.39067750 -0.12235513    -0.20555661 -0.13867505
bar               0.5974211 -0.51506558  0.33396002     0.26041960  0.52297636
pluribus         -0.3396752  0.29972522 -0.26958501    -0.20610932 -0.31033884
sugarpercent      0.1041691 -0.03439296  0.22193335     0.08788927  0.12308135
pricepercent      0.5046754 -0.43096853  0.25432709     0.30915323  0.15319643
winpercent        0.6365167 -0.38093814  0.21341630     0.40619220  0.19937530
                 crispedricewafer        hard         bar    pluribus
chocolate             0.34120978 -0.34417691  0.59742114 -0.33967519
fruity               -0.26936712  0.39067750 -0.51506558  0.29972522
caramel               0.21311310 -0.12235513  0.33396002 -0.26958501
peanutyalmondy       -0.01764631 -0.20555661  0.26041960 -0.20610932
nougat               -0.08974359 -0.13867505  0.52297636 -0.31033884
crispedricewafer      1.00000000 -0.13867505  0.42375093 -0.22469338
hard                 -0.13867505  1.00000000 -0.26516504  0.01453172
bar                   0.42375093 -0.26516504  1.00000000 -0.59340892
pluribus             -0.22469338  0.01453172 -0.59340892  1.00000000
sugarpercent          0.06994969  0.09180975  0.09998516  0.04552282
```

```
pricepercent          0.32826539 -0.24436534  0.51840654 -0.22079363
winpercent            0.32467965 -0.31038158  0.42992933 -0.24744787
                 sugarpercent pricepercent winpercent
chocolate          0.10416906    0.5046754  0.6365167
fruity            -0.03439296   -0.4309685 -0.3809381
caramel            0.22193335    0.2543271  0.2134163
peanutyalmondy     0.08788927    0.3091532  0.4061922
nougat             0.12308135    0.1531964  0.1993753
crispedricewafer   0.06994969    0.3282654  0.3246797
hard               0.09180975   -0.2443653 -0.3103816
bar                0.09998516    0.5184065  0.4299293
pluribus           0.04552282   -0.2207936 -0.2474479
sugarpercent       1.00000000    0.3297064  0.2291507
pricepercent       0.32970639    1.0000000  0.3453254
winpercent         0.22915066    0.3453254  1.0000000
```

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
corrplot(cij)
```



> **Q22.** Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and chocolate are anti-correlated.

> **Q23.** Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I expect fruity and chocolate to have the largest contributions.

Principal Component Analysis

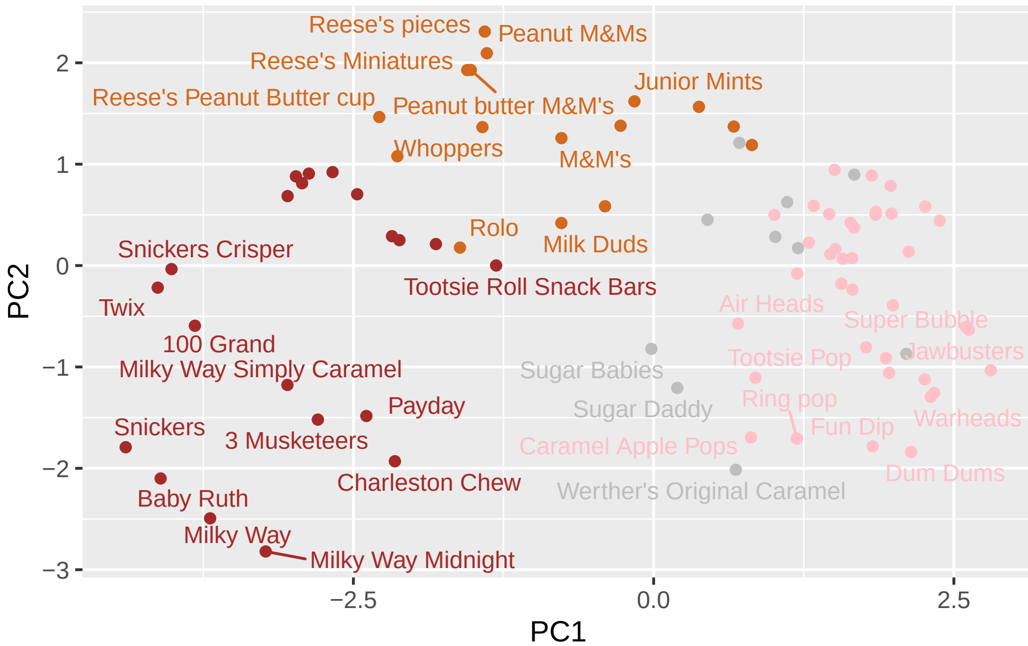
In this case, we want to be sure to set 'scale=TRUE' argument for 'prcomp()' because we have one variable 'winpercent' that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

First major result figure is the "score plot" of PC1 and PC2 – how different candy are related to each other on our new PC axis:

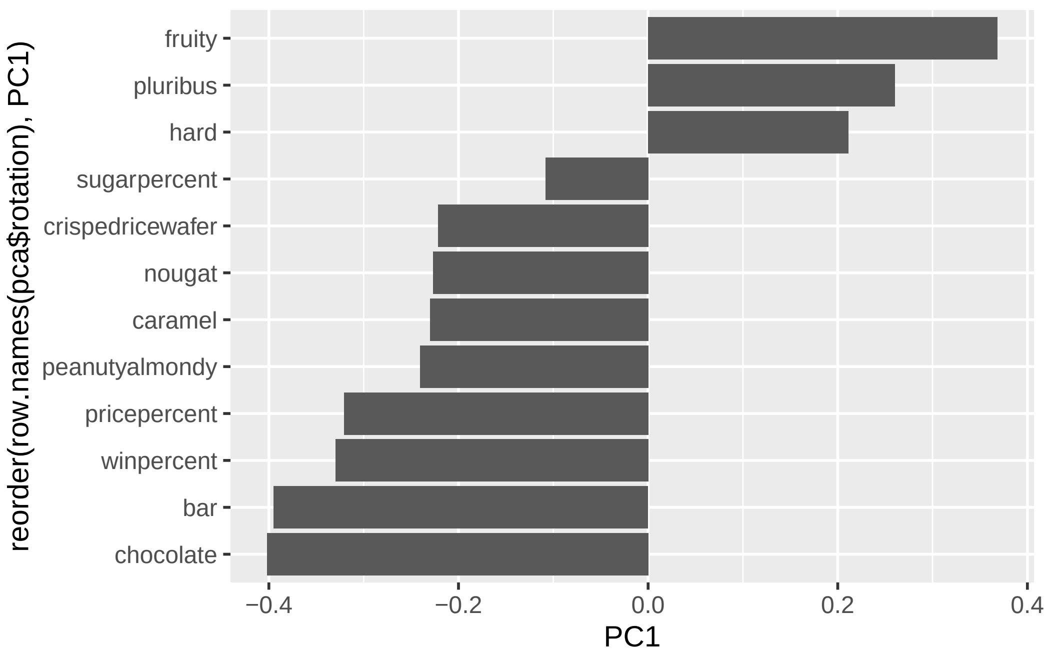
```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=mycols) +
  geom_text_repel(size=3.3, col=mycols, max.overlaps=7)
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The second major results figure from PCA.

```
ggplot(pca$rotation) +
  aes(PC1,
      reorder(row.names(pca$rotation), PC1)) +
      geom_col()
```



Fruity, pluribus, and hard are picked up strongly by PC1 in the positive direction. It makes sense as in the previous figure, fruity candies are on the far right (positive direction) of PC1.

Summary

> **Q25.** Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The characteristics that appear to make a "winning" candy are chocolate and bar, not fruity candies. The visualization figure shows that they are the highest ranked candies. The correlation plot shows that chocolate and fruity are negatively correlated with each other. The PCA plot shows that chocolate and fruity candies tend to be on opposite sides of PC1.